

Reduced-model theory advances — synthesis

2026-06-25

Reduced-model theory advances — synthesis (2026-06-22)

A parallel theory pass that pushed each of the six tractable tempotron reductions (../INDEX.md) farther, grounded in the newly-swept sources (../data-sources/INDEX.md). Six statmech-theory agents produced one derivation each; three independent **codex (gpt-5.5, xhigh)** cross-checks then attacked the hardest open frontiers. Codex did not merely confirm — it **refuted two assumptions baked into the program** and **closed one open derivation**. Full per-model write-ups in this directory; codex reasoning traces in `codex-crosschecks/`.

Honesty note (per ../../CLAUDE.md): every result below is **PARTIAL** — solid mechanism, explicit obstruction. Nothing here is a finished capacity theorem. The value is sharper predictions and the elimination of two wrong premises.

The unifying picture

Every reduction reduces to the same skeleton: capacity is set by an **effective number of independent readouts K_{eff}** , entering an extreme-value lift

$$\alpha_c \approx (\ln \ln K_{\text{eff}}) / (2 \ln 2) + [\text{subleading}],$$

and the lift **is** the log of the solution-cluster count — fragmentation and capacity are one effect counted twice (#3). The models differ only in what sets K_{eff} :

#	Model	What sets K_{eff}	Headline advance	Status
1	discrete binned	the bin count K	$\alpha_c(K)$ lift CONFIRMED: $\hat{\alpha}_c$ 2.5→4.15 ($K=2 \rightarrow 64$); linear rejected, $\ln \ln K$ favored over $\ln K$ ($\Delta\text{AIC} \approx 3.6$, not decisive); slope 0.97	TESTED [OK]
2	spherical (null)	$K=1$	$\alpha_c^+(\kappa) = \frac{1}{2} \cdot \alpha_c^{\text{unc}}(\kappa)$ CONFIRMED $\forall \kappa$ (ratio 0.497±0.021); $p_{\text{silent}}=0.50$ $\forall \kappa$	TESTED [OK]
3	EVT max-of-fields	K correlated fields	shattering CONFIRMED: $q_0(K)$ 1.0→0.14 toward $1/\sqrt{N}$ floor; $\rho=1$ control flat	TESTED [OK]
4	GP / frozen-kernel	kernel spectrum	saturation CONFIRMED: $E[\max V]$ plateaus@OU 2.65 while up-crossings diverge; capacity plateaus	TESTED [OK]

#	Model	What sets K_{eff}	Headline advance	Status
5	soft-max (β)	K , dialed by β	β-knob CONFIRMED (qual): $\hat{\alpha}_c$ 2.13→2.43, $C \approx 0.51$, sub-quadratic past β^* , ρ -suppression $1.9 \times$	TESTED [OK]
6	sinusoidal	bandwidth $\times T$	$K_{\text{eff}} = \Omega T / \sqrt{3}$ exact; $E[N_{\text{max}}] = (T/2\pi)\sqrt{(\lambda_4/\lambda_2)}$ exact; capacity LIFT measured on the certified axis ($\hat{\alpha}_c$ 2.45→2.75 for K_{eff} 8.6→61, slope ≈ 0.45)	PARTIAL→ tested
7	cross-kernel universal-ity	bandwidth $\times T$ and shape	capacity is TWO-AXIS, NOT $f(K_{\text{eff}})$ alone: band-pass (gabor) > low-pass (gauss) at matched K_{eff} (+0.3-0.6), reaching $\hat{\alpha}_c \approx 3.9$ near the radial bound 4; slope $d\hat{\alpha}_c/d\ln K$ falls 0.95→0.74→0 with Q ; sinusoid flat ≈ 3.4	TESTED [OK]

What codex changed (independent cross-checks)

1. There is no finite α_0 constant (codex-r1). The first-moment/cluster calculation gives

$$\alpha_c(K) = (\ln \ln K)/(2 \ln 2) + (\ln \ln \ln K)/\ln 2 - 0.383... + o(1),$$

so the residual $\alpha_c - (\ln \ln K)/(2 \ln 2)$ does **not** converge — the next term is a third iterated log, not a constant. The “ $\alpha_0 \approx 2.58$ ” carried through #3/#4/#6 (and the program’s fits) is a **small-K fit artifact**, not an asymptotic constant. Status: the bivariate EVT step is rigorous; the cluster step is Rubin-level heuristic and a true subleading answer needs a controlled quenched/RSB calculation. **Action taken:** the “ $+\alpha_0 \approx 2.58$ ” in #3/#4/#6 is now flagged as a fit, not a derived constant.

2. The #1 $\sqrt{K/N}$ finite-N “starvation” law is not supported in the ideal model (codex-r1). In the exact Gaussian max-of-fields reduction the fields are exactly Gaussian and bins exactly independent, so the max has **no leading finite-sample bias**: the $\sqrt{K/N}$ and K/N coefficients are **zero**, and finite-N corrections are ordinary saddle corrections $\delta(K,N) = C_K/N + o(1/N)$. Consequently the empirical rise $\hat{\alpha}_c \approx 3.0$ ($N=200$) → 3.4–3.5 ($N=500$ –1000) is **not** explained by clean finite-N theory (would need $C_K \sim 100$). It must come from outside the idealization: the **$K=O(N)$ density regime** (the program’s own $2N/K$ finding — [[tempotron-capacity-finite-N-starvation]]), time-discretization, or algorithmic/non-Gaussian effects. This **supersedes** the #1 derivation’s $\sqrt{K/N}$ headline.

3. The #4 saturation prediction is independently confirmed (codex-m4). As $\tau_s \rightarrow 0$ at fixed T, τ the membrane process converges (in $C[0, T]$) to an Ornstein-Uhlenbeck process, whose extreme value obeys Berman/Pickands asymptotics $E[\max] = \sqrt{2 \ln(T/\tau)} + o(\sqrt{\ln})$ — the **participation-ratio** scale $K_{\text{eff}} \sim T/\tau$, **not** the up-crossing count $K_{\text{up}} = T/\sqrt{(\tau\tau_s)}$. The divergent high-frequency up-crossings carry vanishing power and add no lift. Two independent derivations now agree: **capacity saturates as the synaptic rise sharpens**.

4. The #5 finite- β profile is now closed at small β (codex-m5):

$$\alpha_c(\beta) = 2 + [(K-1)(1-\rho)^2] / [K(1+(K-1)\rho)] \cdot \beta^2 + O(\beta^3) \text{ (balanced labels, fixed } K, \beta \ll 1),$$

with explicit $K=2,3$ constants; $\beta \rightarrow 0 \rightarrow \alpha_c = 2$ verified. The Gaussian distance-reduction is rigorous; the capacity is RS-level. This complements the agent's fragmentation onset $\beta^* \sim \sqrt{2 \ln K}$: the learning surface stays convex with a **quadratic-in- β capacity rise** until β^* , then enters the shattered (RSB) regime.

5. The #2 κ -dependent silent-fraction conjecture is refuted; the halving is uniform (codex-m2). The excitatory-only ($\omega \geq 0$) spherical perceptron has $\alpha_c^+(\kappa) = 1/[2 I_\kappa] = \frac{1}{2} \cdot \alpha_c^{\text{unc}}(\kappa)$ **exactly for all margins κ** , with $I_\kappa = (1+\kappa^2)\Phi(\kappa) + \kappa\phi(\kappa)$, and silent fraction **$p_{\text{silent}} = \frac{1}{2}$ at every κ** (the conjectured nontrivial $p_{\text{act}}(\kappa) = 1 - H(b(\kappa))$ is wrong — the saddle has $b=0$). RS is stable on the convex positive-margin cone (no RSB), and the result is independently confirmed by a non-replica **CGMT/Gordon** argument — the most rigorous result of the pass. So the biologically-faithful null is a clean uniform halving of the unconstrained Gardner curve.

Sharp, falsifiable predictions (for the lab, not run here)

- 1. Rise-time saturation (strongest).** Simulate the exact stationary GP from the double-exponential PSD at fixed T, τ while $\tau_s \rightarrow 0$: **$E[\max V(t)]$ plateaus** while the raw up-crossing count diverges. Two independent derivations (#4 + codex OU limit).
- 2. Kernel-tuned capacity.** α_c is maximized at well-separated τ , τ_s ($K_{\text{eff}}^{\text{PR}}$ peaks), not at $\tau_s \rightarrow 0$. Directly contradicts the up-crossing intuition.
- 3. β -profile.** $\alpha_c(\beta) - 2 \propto \beta^2$ at small β with the K, ρ -explicit coefficient above; fragmentation/RSB onset near $\beta^* \sim \sqrt{2 \ln K}$.
- 4. Finite-N is density, not size.** At fixed K , the ideal model predicts only an $O(1/N)$ correction; the observed large N -rise is the $K=O(N)$ density ($2N/K$) regime leaving the asymptotic window — consistent with the program's single-learner scaling.
- 5. Sinusoidal exactness.** $K_{\text{eff}} = \Omega T / \sqrt{3}$ (flat band) and $E[N_{\text{max}}] = (T/2\pi) \sqrt{(\lambda_4/\lambda_2)}$ are exact, finite-N-checkable witnesses (#6, MC-verified).

Lab campaign results (2026-06-22→23) — ALL SIX reductions tested

Every reduction was taken to the lab (the laboratory skill; ~\$42 total, each model under its \$25 cap). **6/6 predictions CONFIRMED.** Per-model records in `experiments/<name>/EXPERIMENT.md`; each model's academic PDF now carries an Experimental Test section; pipeline in `CAPACITY-EXPERIMENT-PLAYBOOK`

- #2 spherical (null) — CONFIRMED, \$0.04.** Convex LP/QP existence: $\alpha_c^+(\kappa)/\alpha_c^{\text{unc}}(\kappa) = 0.497 \pm 0.021$ flat $\forall \kappa$ (exact halving); $p_{\text{silent}} = 0.50$ uniform $\forall \kappa$. The cleanest result (CGMT-exact, no learner).
- #4 GP/frozen-kernel — CONFIRMED, \$0.92.** Free GP-sim witness: $E[\max V]$ **plateaus** at the OU value 2.65 ($=\sqrt{2 \ln K_{\text{PR}}}$) while up-crossings **diverge** $\propto 1/\sqrt{\tau_s}$; findable capacity plateaus. Sharpening synapses does NOT buy unbounded capacity — it's the participation ratio, not the crossing count.
- #5 soft-max(β) — CONFIRMED (qual), \$16.4.** $\hat{\alpha}_c$ rises 2.13→2.43 with β , $C \approx 0.51$ (vs existence 0.667, findable < existence), sub-quadratic past β^* , ρ -suppression 1.9 $\times$. Priciest: the smooth learner is slow.
- #1 discrete max-of-K — CONFIRMED, \$5.** $\hat{\alpha}_c(K)$ 2.5→4.15 ($K=2 \rightarrow 64$); **linear growth decisively rejected; $\ln \ln K$ favored over $\ln K$** ($\Delta \text{AIC} \approx 3.6$ — positive but not decisive; the wide direct-K span makes the discrimination possible), slope 0.97 (subleading-steepened).
- #3 EVT max-of-fields — CONFIRMED, \$1.** Solution overlap $q_0(K)$ **1.0→0.14** toward the $1/\sqrt{N}$ floor (shattering); $\rho=1$ control flat at 1. On #1's runs: **the capacity lift IS log(cluster count), counted twice** — direct evidence for the shattered-learning-surface thesis.

#6 sinusoidal capacity (the first run)

Prediction 5 was **tested** (\$11 of \$25 budget). On the certified $K_{\text{eff}} = \Omega T / \sqrt{3}$ axis (witness-verified $\leq 4\%$), the band-limited tempotron shows a genuine **temporal capacity lift above the perceptron $\alpha_c = 2$** : findable $\hat{\alpha}_c = 2.45, 2.55, 2.65, 2.75$ at $K_{\text{eff}} \approx 8.6, 15.6, 31.1, 60.6$ (fixed input density), measured $\ln \ln K_{\text{eff}}$ slope ≈ 0.45 (below the EVT existence slope 0.721 — the expected findable < existence gap). A density-control sweep saturates, so the lift is not a density artifact. **Process note (important)**: the first pass was a compound artifact — an under-budgeted/unvalidated learner (false flat ~ 2.1 ceiling) plus a density confound (mean_spikes $\propto K_{\text{eff}}$); both were caught by plotting run metrics (epochs to solve, learning curves) and fixed by holding density constant + validating the learner via critical slowing. The growth law ($\ln \ln K$ vs $\ln K$) and the asymptotic 0.721 are NOT pinned (K range 8–64 too narrow). Full record: experiments/sinusoidal_capacity/EXPERIMENT.md; paper pdf/06-sinusoidal-kernel.pdf.

Finite-N ladders (2026-06-24) — an honest course-correction

A dedicated finite-N campaign varied N at fixed model-knob for all six reductions (predictions set by a naive $\rightarrow$ statmech $\rightarrow$ codex **tiered cascade**, finite-N-prediction-synthesis.md; each result numerics-verifier significance-gated). **The program's prior headline — findable slope rises with N toward 0.721 (masking $\rightarrow$ recovery) — does NOT survive once the learner is budget-validated and density is fixed.** With those controlled, the ideal/findable **per-cell capacity is N -flat**, and the $\ln \ln K$ slope is a finite- K -window property (above the asymptotic 0.721 via the $(\ln \ln \ln K) / \ln 2$ subleading), not an N -recovery. The earlier “rises” were **learner-budget + density confounds**.

- **#1 discrete**: per-cell $\hat{\alpha}_c$ N -flat ($K = 4/16/64 \times N = 250/500/1000, \pm 0.15$); window slope 1.1–1.3, N -independent, > 0.721 . The old “finite-size rise 3.06 $\rightarrow$ 3.49” = the **$K = O(N)$ density regime**, not size.
- **#6 sinusoidal**: with a validated learner, per-cell N -flat, slope $\sim 0.9 \rightarrow$ **#6 joins #1**. The campaign's 0.45 was an **under-budgeting artifact** (cells need 8k–17k epochs vs the 8000 cap). Masking-recovery with N not resolved (slope Δ within noise).
- **#2 spherical**: ratio flat $0.5 \forall N$; $\alpha_c^{\text{unc}} \rightarrow \text{Gardner} \forall N$; $p_{\text{silent}} \rightarrow 0.5$ (exact; $N \in \{100, 200, 400, 800\}$).
- **#3 EVT**: q_0 **rises** with N (intensive; $K = 8$ 0.26 $\rightarrow$ 0.45) above the falling $1/\sqrt{N}$ floor; $\rho = 1$ ctrl = 1.0 — a genuine N -effect (the floor clearing), confirming q_0 is an intensive order parameter.
- **#5 soft-max**: $C(\beta = 0.5)$ 0.56 $\rightarrow$ 0.665 ($\approx$ existence 0.667) — **consistent with** findability recovery but **within grid resolution (not confirmed)**; $N = 750$ budget-limited.
- **#4 GP**: N/A — the decisive $E[\max V]$ witness is a GP simulation with no weight dimension N .

Process: incremental results.csv writing (salvaged timed-out cells); the significance gate caught two over-claims (#5/#6 “rises” within resolution) and forced calibrated wording. Per-model finite- N sections in each EXPERIMENT.md; PDFs 01–06 updated; details in finite-N-prediction-synthesis.md.

Cross-kernel universality (#7, 2026-06-25) — capacity is two-axis

The sequel to “capacity = $f(K_{\text{eff}})$ ”: is it shape-independent? **No**. Measuring findable $\hat{\alpha}_c$ across a kernel family (gauss, sinc, alpha, gabor at $Q = 2/4/8$, sinusoid) on one certified numeric K_{eff} axis (study v13_kernel_capacity.py; full record experiments/kernel_universality_capacity/EXPERIMENT.md; paper pdf/academic/07-kernel-universality.tex; $\sim \$26$ lab): - ****No single scalar collapses $\hat{\alpha}_c$ **** (offset spread 0.77 on K_{eff} , 0.66 on D_{PR} , 1.17 on K_{eff}/Q) $\rightarrow$ capacity is **TWO-AXIS: effective dimension $\times$ radially/bandwidth**. - **Band-pass beats low-pass at matched K_{eff}** (gabor – gauss = +0.3...+0.6); gabor reaches **** $\hat{\alpha}_c \approx 3.89$ at $K_{\text{eff}} = 64$, near the**

radial/quadratic-perceptron bound $\alpha_c=4^{**}$ (Cover/Gardner). The readout $\max_t w \cdot \varphi(t)$ interpolates linear (broadband, $\rightarrow 2$) $\rightarrow$ radial $\sqrt{(w \cdot a)^2 + (w \cdot b)^2}$ (narrowband, $\rightarrow 4$). - **Decisive slope-vs-Q:** $d\hat{\alpha}_c/d\ln K$ falls 0.95(gauss) $\rightarrow$ 0.74(Q8) $\rightarrow$ 0(sinusoid) — narrowing converts the EVT count-lift into intrinsic radial capacity. The free $E[\max V]$ witness pointed the opposite way (it tracks only the suppressed log count-channel, blind to decision-boundary curvature). - **Honest status:** empirical facts SOLID; the linear $\leftrightarrow$ radial + additive two-channel mechanism is a plausible interpretation (codex flags additivity as the weak link). Decisive follow-up named (match D_{eff} & extremal index, vary only radially). “Capacity = $f(K_{\text{eff}})$ ” holds only within a fixed shape class.

Open items (next derivations)

- **α_0 / iterated-log subleading (#3, all):** needs a controlled quenched/RSB cluster complexity, not the annealed first moment. This is the deepest open problem.
- **#5 finite- β beyond small β :** the $H_\beta(h;q)$ threshold integral over K correlated Gaussians (neither half-space nor product) — the RSB regime above β^* .
- **#1 C_K coefficient:** the one-loop/Hessian correction in the Gaussian reduction.
- **#4 $\bar{K} \rightarrow K_{\text{eff}}^{\text{PR}}$ identification inside the saddle** (currently a physically-motivated hypothesis, not derived from the replica saddle).

Provenance

- Per-model derivations: 01..06-**-advance.md (statmech-theory agents).
- Independent cross-checks: codex-crosschecks/{r1,m4,m5}-*.md (codex gpt-5.5, xhigh).
- Sources: mirrored PDFs in ../../papers/; catalog ../../data-sources/INDEX.md.
- Integrity fix made during this pass: the misfiled Brunel2004 PDF (was Engel-Monasson-Hartmann) was replaced — see commit history.